Figure legends

Figure 1. Closed-corpus retrieval and answer workflow. The archived six-document corpus yielded 382 parsed source records and 481 chunks. The workflow includes glossary rewriting, query expansion, hybrid fusion, article anchors, reranking, conditional sibling expansion, evidence classification and generation. Evaluation includes reused development sets and diagnostic assessments followed by the frozen author-led test. It is not a current-law maintenance system. This depicts the original pipeline; the separately frozen integrated repair and its author-reviewed outcomes are described in Methods and Figure 4. AI assistance: OpenAI Codex assisted plotting or diagram code and presentation from the retained data and documented workflow; source records and code are available for verification. Scientific content and assessment provenance are described in Methods and Supplementary Section S11.

Figure 2. Constructed benchmark design and archived source coverage. The benchmark has 140 direct-article questions, 20 multi-article or exception questions and 40 unanswerable or adversarial questions, organized as 100 bilingual pairs. The lower panel counts each source once per question: 28 each for RDC 9/2015, RDC 55/2010, RDC 751/2022 and RDC 406/2020, 26 for RDC 945/2024 and 22 for IN 338/2024. The former count of 30 included duplicate source/article entries. RDC 9/2015 is revoked. Coverage is not applicable-law correctness. AI assistance: OpenAI Codex assisted plotting or diagram code and presentation from the retained data and documented workflow; source records and code are available for verification. Scientific content and assessment provenance are described in Methods and Supplementary Section S11.

Figure 3. Full-task success in the frozen 40-answer author-led assessment. Both languages must pass for a pair to succeed. Counts are 6/12 answerable, 0/4 partly answerable, 4/4 unanswerable and 10/20 overall. Error bars are exact two-sided 95% binomial intervals at pair level. They are conditional on this selected sample and do not account for author or topic dependence. AI assistance: OpenAI Codex assisted plotting or diagram code and presentation from the retained data and documented workflow; source records and code are available for verification. Scientific content and assessment provenance are described in Methods and Supplementary Section S11.

Figure 4. Completed author-reviewed outcomes in the exploratory 80-answer comparison. Panel A shows baseline and repaired pair success overall and by answerability category; both languages must meet the task criteria, and error bars are exact two-sided 95% binomial intervals. Overall counts are 14/20 and 17/20, answerable 10/12 and 11/12, partly answerable 0/4 and 2/4, and unanswerable 4/4 in each condition. Panel B retains the pairing: 13 pairs passed both conditions, four improved, one worsened and two failed both. The paired difference is +15.0 percentage points (95% pair-bootstrap interval -5.0 to +35.0; exact two-sided discordant-pair $p = 0.375$). The comparison is inconclusive. AI-prepared rating assistance, differing author exposure and selected archived tasks limit inference. This figure presents the completed author-review overlay, not the earlier AI scores, independent external validation or current-law accuracy. AI assistance: OpenAI Codex assisted plotting or diagram code and presentation from the retained data and documented workflow; source records and code are available for verification. Scientific content and assessment provenance are described in Methods and Supplementary Section S11.

Figure 5. Coauthor assessed answer comparison and matched retrieval baseline. Panels A to F use the same 40 language-specific questions per condition, grouped into 20 bilingual pairs. Hybrid passages, BM25 passages and model-only answers use a common prompt; model-only receives no passages or web search. A shows content and evidence-boundary success, including appropriate abstention on eight absent-detail answers per condition. B and C partition those questions by topic and archived support, not online availability. D shows affected answers with omissions or material incorrect/unsupported claims. E shows author-rated article-citation presence; citation support among citing answers is 39/39 for hybrid and 37/38 for BM25, with no model-only citations. Author and AI citation judgments differ and are retained separately. F shows correction flags, not measured human effort. G compares structural retrieval coverage on 32 positive-target questions, excluding eight absent-detail questions, with the same archived chunks, saved queries, five seed chunks and completion policy. The small differences do not establish hybrid superiority or equivalence. BM25 answer generation is sequential and reuses saved hybrid comparators. Human assessment is by a coauthor with prior project exposure and limited condition masking; it is not independent external validation. AI assistance: OpenAI Codex assisted plotting or diagram code and presentation from the retained data and documented workflow; source records and code are available for verification. Scientific content and assessment provenance are described in Methods and Supplementary Section S11.

Supplementary Figure S1. Task topics and coauthor assessed answer outcomes. Each row is one of 20 bilingual pairs; columns show Portuguese and English answers for hybrid, BM25 and model-only conditions. S denotes content success with supported information; A, appropriate abstention for an absent-detail question; H, a material incorrect/unsupported claim flag, whether or not the answer is complete; I, another content failure. H takes precedence over I. These derived display classes summarize the returned ratings and do not replace individual criteria. All six outcomes use the same task reference; no web search was used. Archived support does not describe online availability. AI assistance: OpenAI Codex assisted plotting or diagram code and presentation from the retained data and documented workflow; source records and code are available for verification. Scientific content and assessment provenance are described in Methods and Supplementary Section S11.

Figure 6. Dated corpus comparison with provisional AI assessment. Panels A, C and D use the same 188 language-specific questions per condition; B requires both languages to pass within each of 94 pairs. The historical repaired corpus and updated consolidated corpus shared downstream settings and approved Portuguese retrieval queries. A shows 144/188 versus 172/188 primary successes, with a paired difference of 14.9 percentage points (descriptive 95% bilingual-pair bootstrap interval 7.4 to 22.9; shared-article-family sensitivity 7.1 to 23.1). B shows 69/94 versus 83/94 successful pairs. C partitions all 188 questions into six prespecified topics with equal denominators across conditions. D shows overlapping omission, material-error, source-support, applicability, language and correction flags; correction flags do not measure human editing time. Substantive success excludes exact-label citation-format normalization. The 376 records derive from 366 distinct prompts and ten exact cross-condition reuses. Source updating and structural curation are bundled; newly available sources are separated in Supplementary Table S23. All new output ratings are provisional Codex assessments with development involvement, not new human verification or independent validation. Supplementary Data S16 contains source-linked counts and initial ratings. AI assistance: OpenAI Codex assisted analysis and plotting from retained records.